

Web Programming

Web Programming

Rodrigo Martins Pagliares

Web programming foundations

The web development career

Types of web developers and the need for T-shaped individuals

Lecture: Types of web developers and the need for T-shaped individuals
Module: The web development career

Types of web developers

- **Typical job titles in web development** are divided in **three** different categories:
 1. **Front-end** developers.
 2. **Back-end** developers.
 3. **Full-stack** developers.

Front-end developer

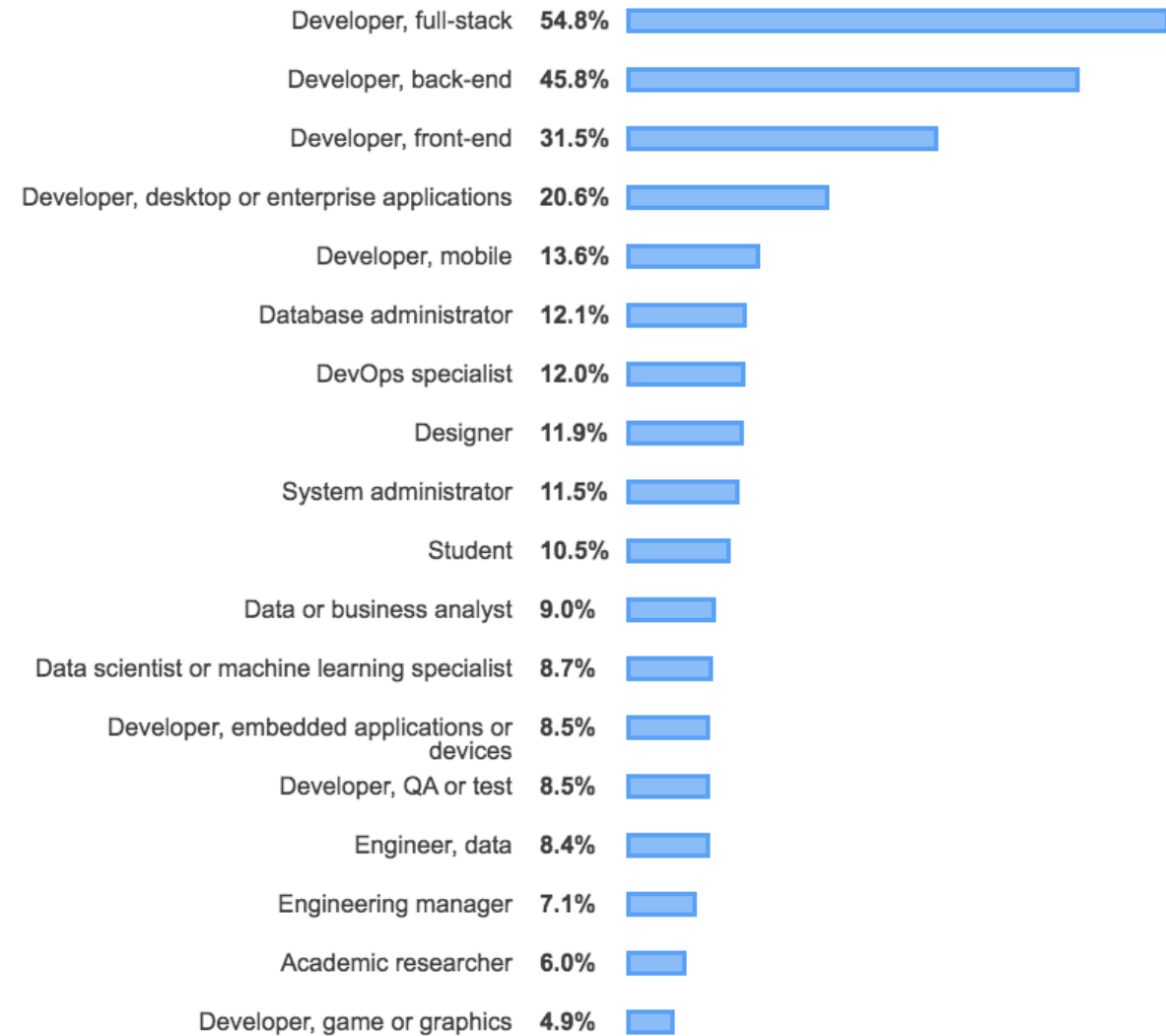
Front-end developer

- Focuses on what's in front of a user.
- Usually some type of web browser like:
 - ✓ Apple's Safari
 - ✓ Mozilla's Firefox
 - ✓ Google's Chrome
 - ✓ Microsoft's Edge
 - ✓ ...
 - ✓ ...



Front-end developer

- About 31% of web developers in United States are front-end developers that deal with code that manages the interaction between the user and the browser.



Source: <https://insights.stackoverflow.com/survey/2019>

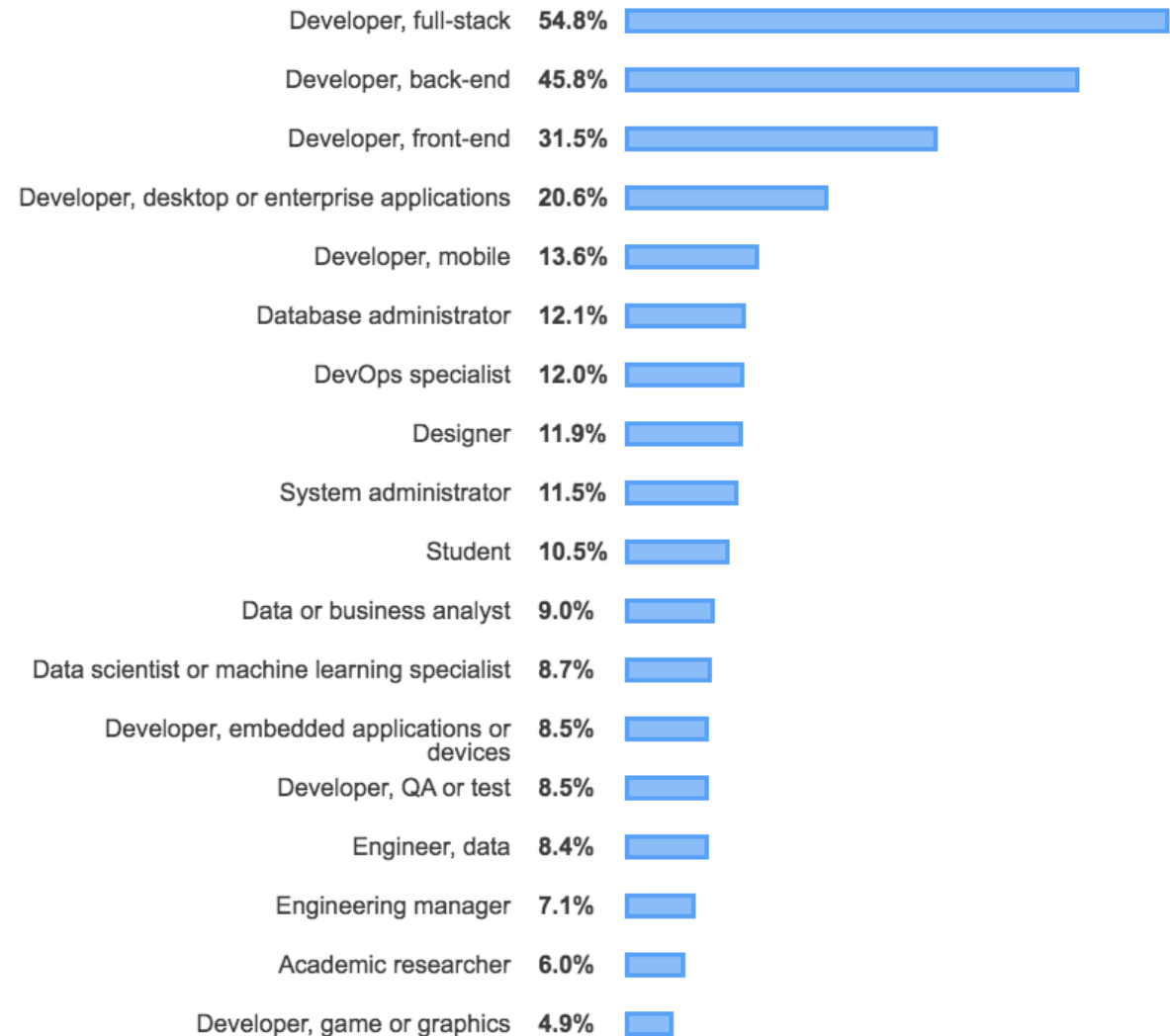
Front-end developer

- A **front-end developer** works in the **design** of the **page** and in the **interactions between users and websites**.
- As **market** has **changed over the years**, **front-end developers** are expected to **know** more about **interfacing with servers**, as well as **programming**.

Back-end developer

Back-end developer

- About 46% of all developers identify themselves as back-end developers.
 - ✓ Focus on what happens behind the scenes on a server.
 - ✓ Deal with setting up server machines, designing the structure of the information, and managing the apps and services that users communicate with.

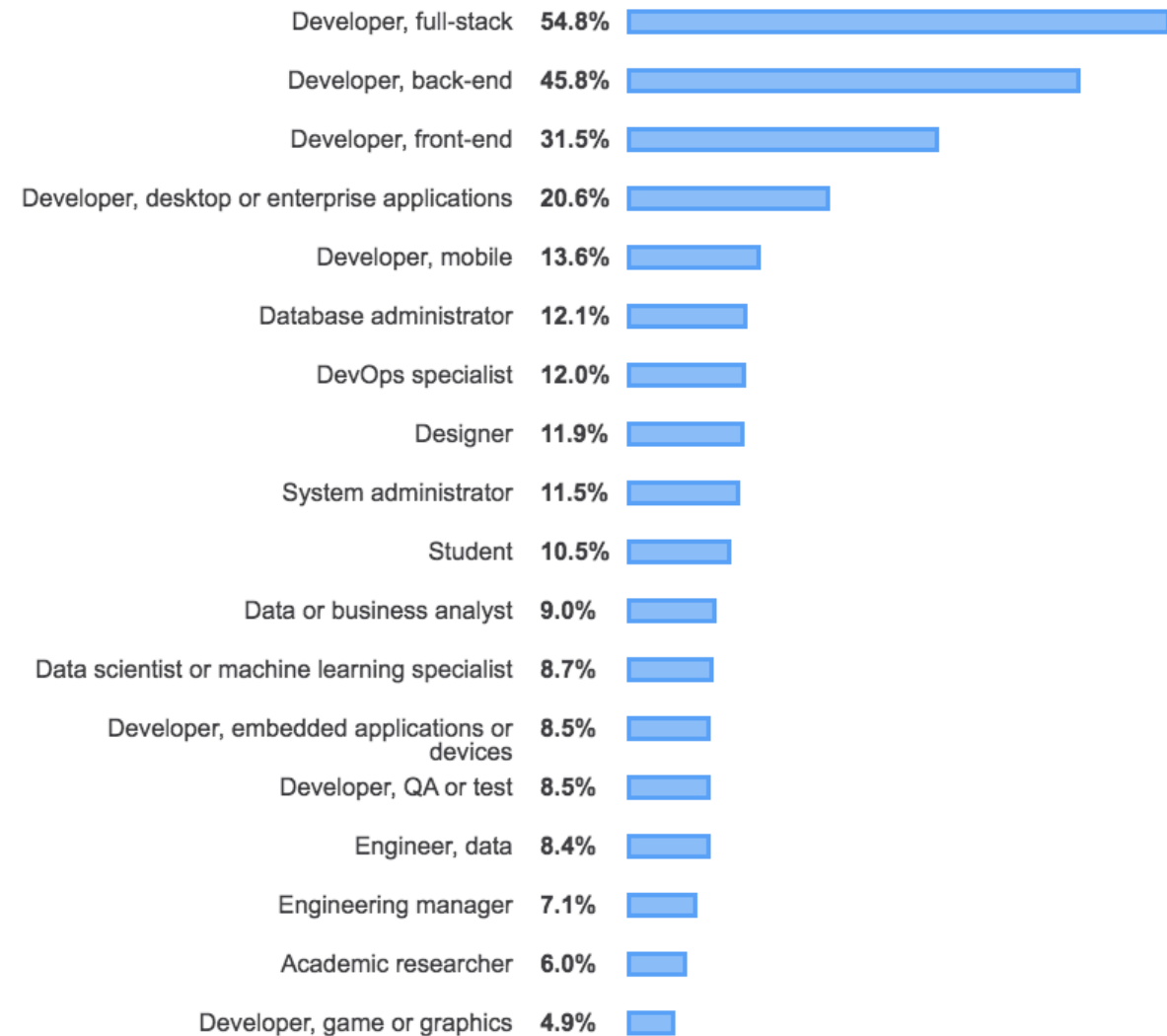


Source: <https://insights.stackoverflow.com/survey/2019>

Full-stack developer

Full stack developer

- Most developers identify themselves as **full-stack developers**.
 - ✓ Combination of a front-end developer and a back-end developer.
 - ✓ Someone that can handle both the front and the back-end of web development.



Source: <https://insights.stackoverflow.com/survey/2019>

Full stack developer

- In reality, there's no perfect front-end or back-end developer.
 - ✓ All developers are somewhere in between.
- The notion of a **full-stack developer**, is somewhat of a **unicorn**.





T-shaped individuals

T-shaped individuals



- **Developers need to have a t-shaped strategy for learning.**
- **Developers tend to learn a wide variety of skills, but specialize in a few that they enjoy.**
 - ✓ It's also smart to specialize in what makes you more marketable.
- That's **why** it's important to **keep learning**, because **what's marketable changes constantly**.

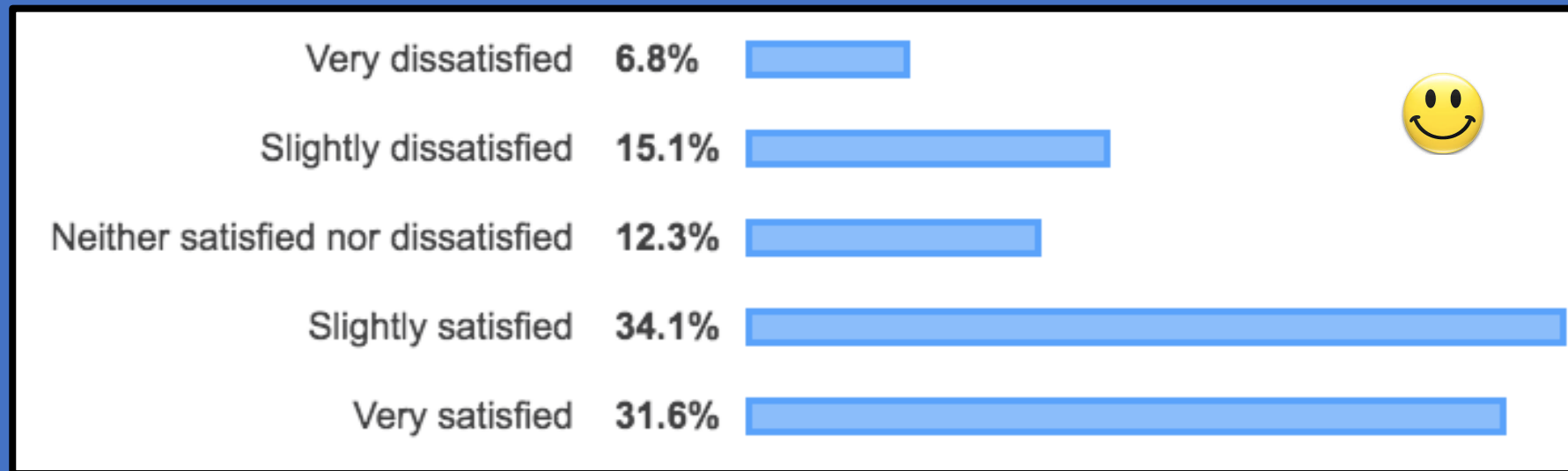
Job satisfaction and manager confidence

Lecture: Job satisfaction and manager confidence

Module: The web development career

Job satisfaction

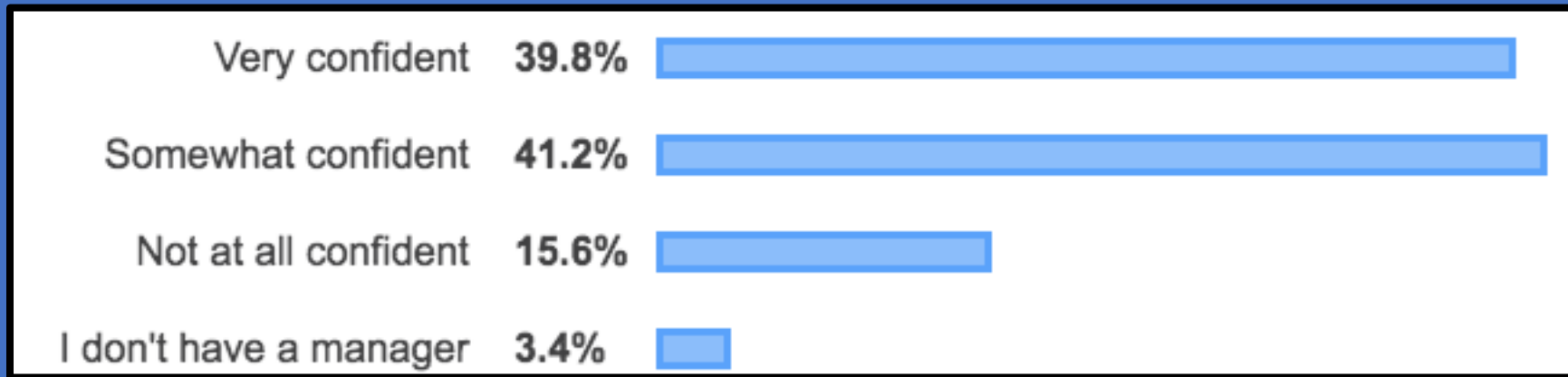
- Overall, developers are **satisfied** with their jobs.
 - ✓ Almost **66%** are **satisfied** with their **current position**.



Source: <https://insights.stackoverflow.com/survey/2019>

Manager confidence

- Over **80%** of **respondents** are **confident** that their **manager knows what they're doing**.



Source: <https://insights.stackoverflow.com/survey/2019>

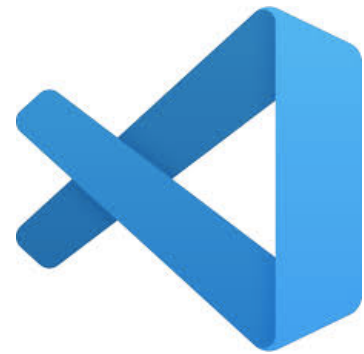
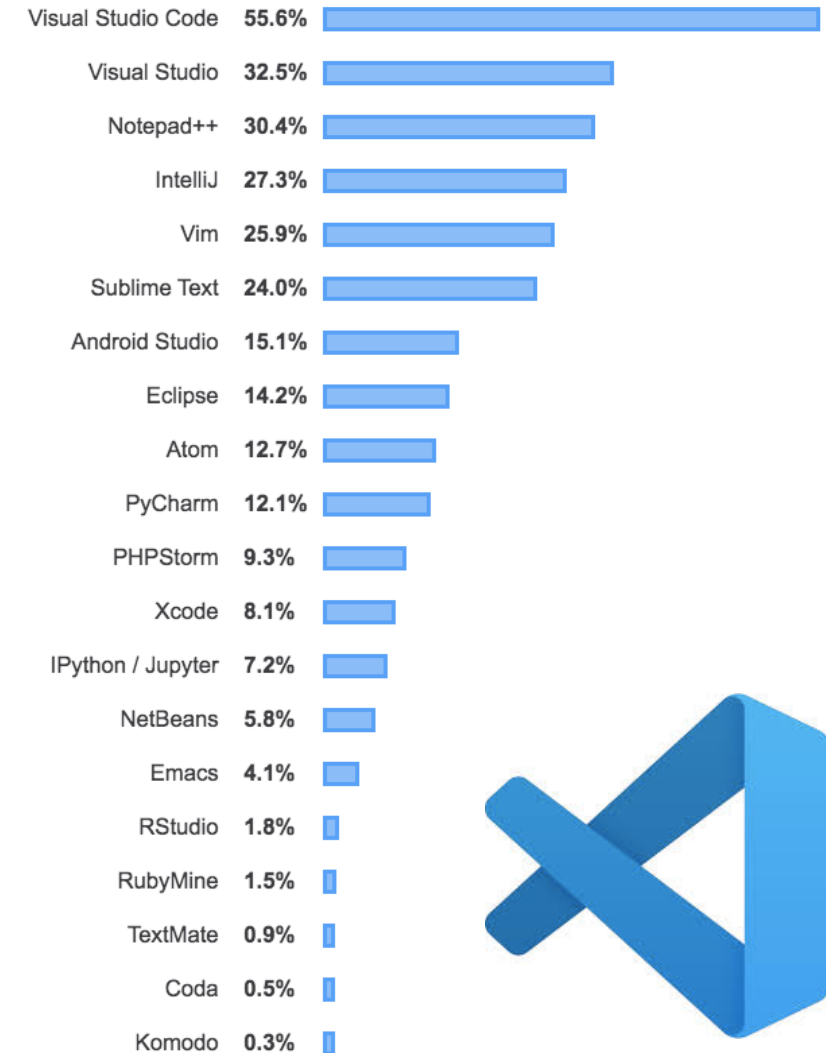
Web development

Lecture: Web development

Module: The web development career

Development environments

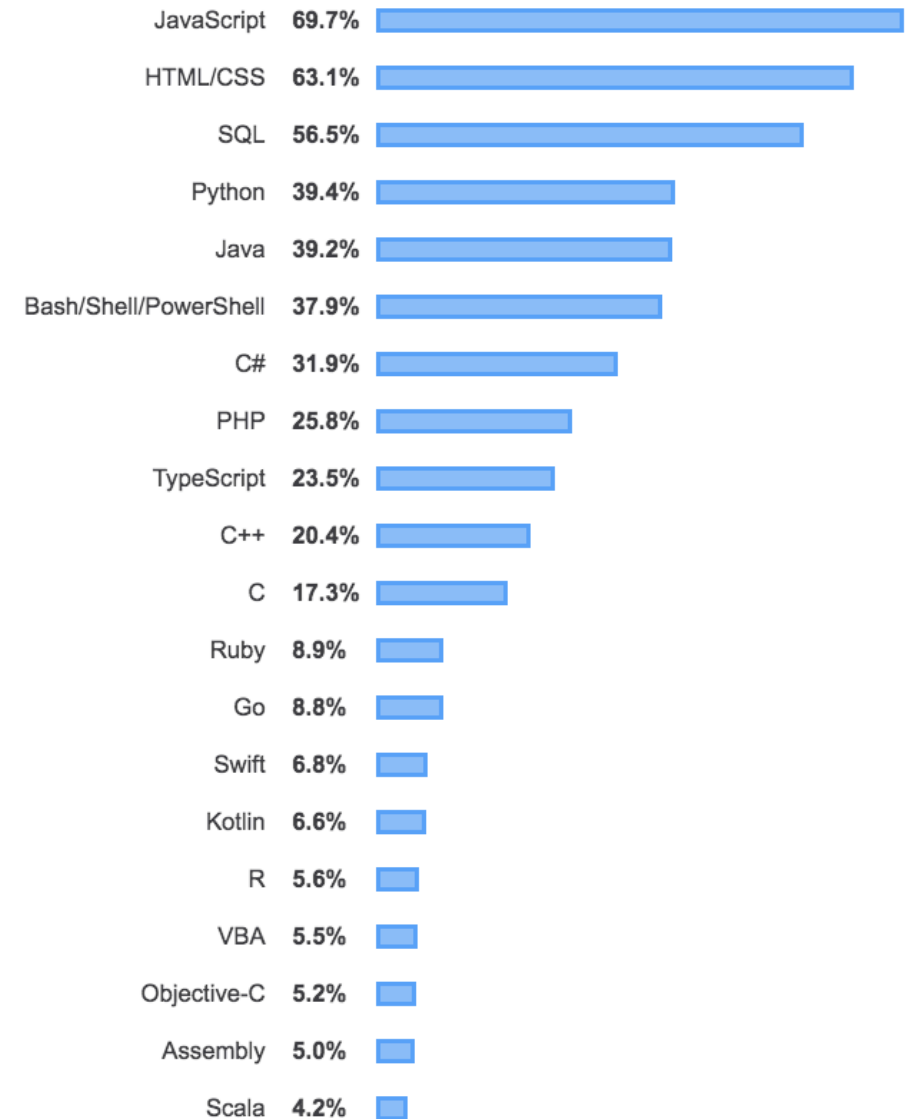
- Visual Studio Code is a dominant player among developer environment tools in 2019.



Source: <https://insights.stackoverflow.com/survey/2019>

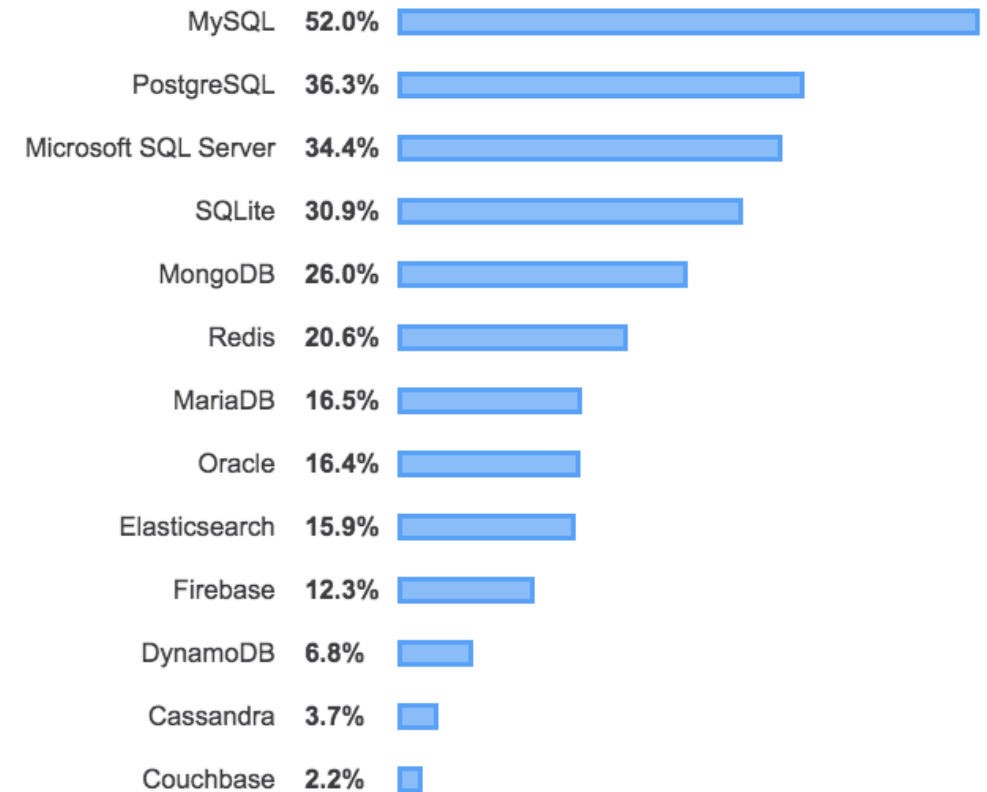
Programming, scripting and markup languages

- For the seventh year in a row, **JavaScript** is the most commonly used **programming language**.
 - ✓ **Python** has risen in the ranks *again*.
 - It has just **edged out Java** in overall ranking.



Database systems

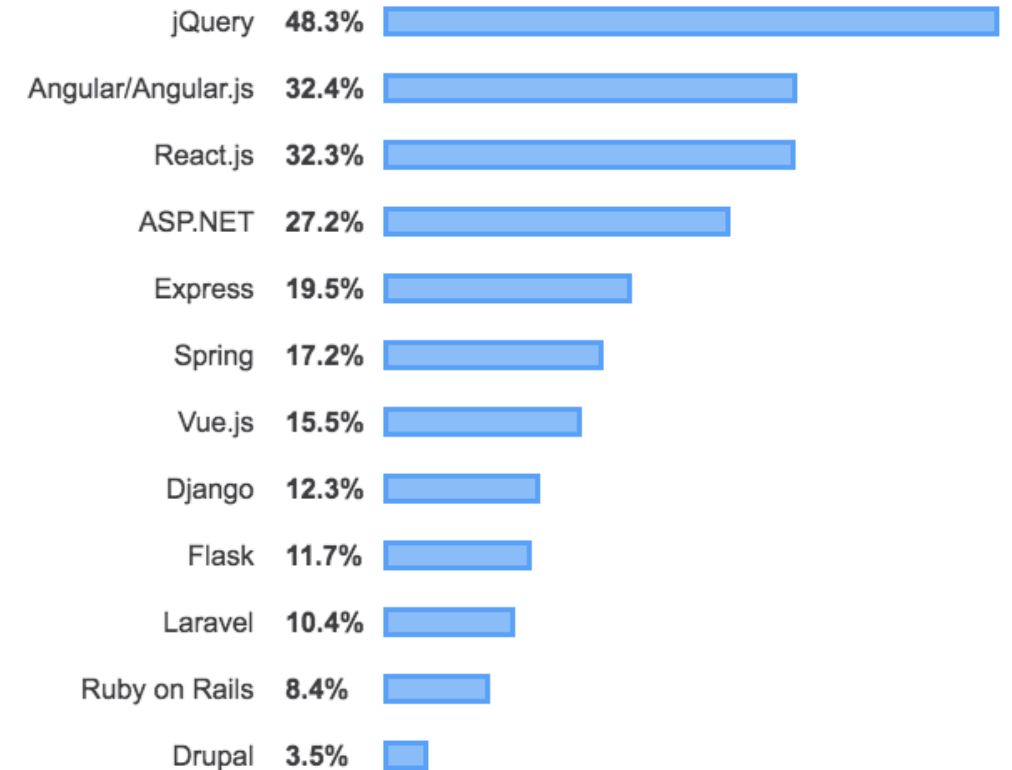
- **MySQL** is the most commonly used database system.
- **PostgreSQL** has taken the **second spot** this year, edging ahead of **Microsoft SQL Server**.



Source: <https://insights.stackoverflow.com/survey/2019>

Web frameworks

- **jQuery** is the most **broadly used** web framework



Source: <https://insights.stackoverflow.com/survey/2019>

**Developers have to learn to use different types
of languages and development environments
that change between projects.**

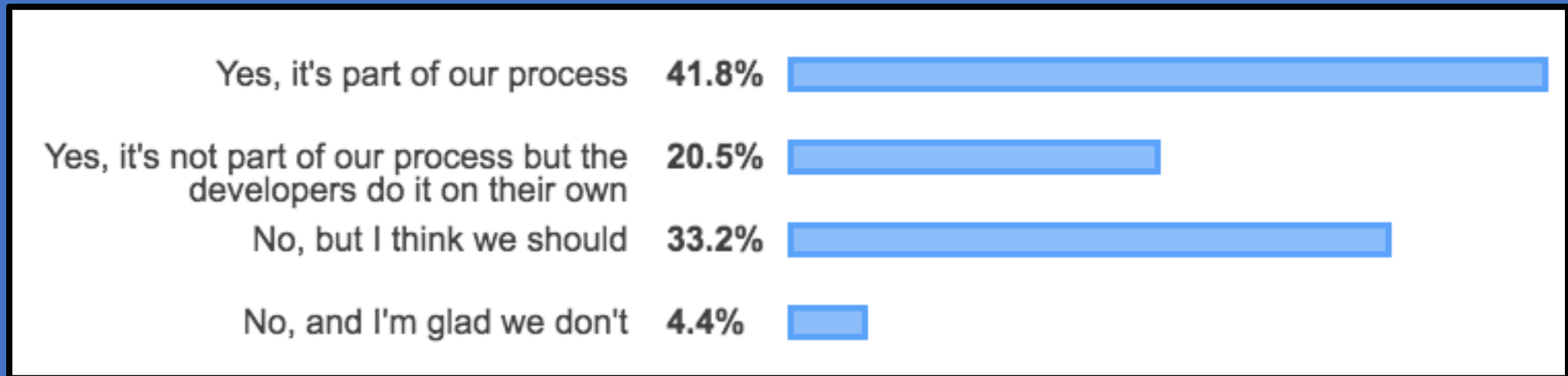
Agile practices and contribution to open-source projects

Lecture: Agile practices and contribution to open-source projects

Module: The web development career

Unit tests

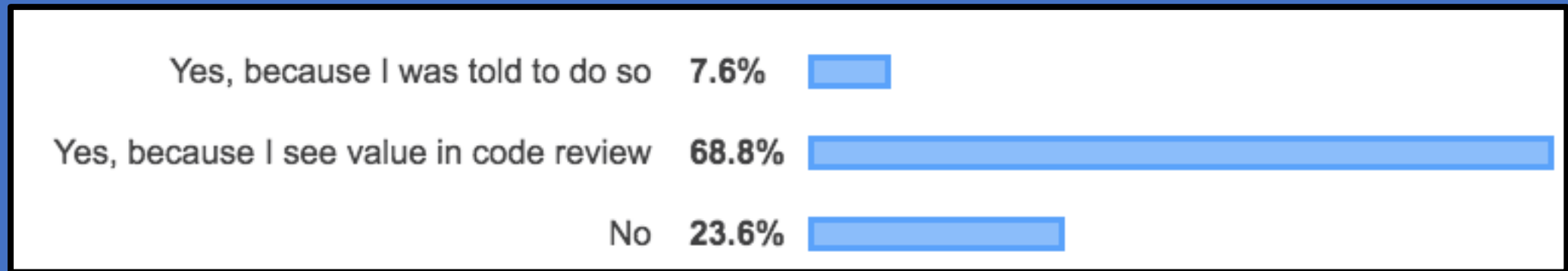
- **Developers at companies who embrace unit testing also have slightly higher job satisfaction.**



Source: <https://insights.stackoverflow.com/survey/2019>

Code review

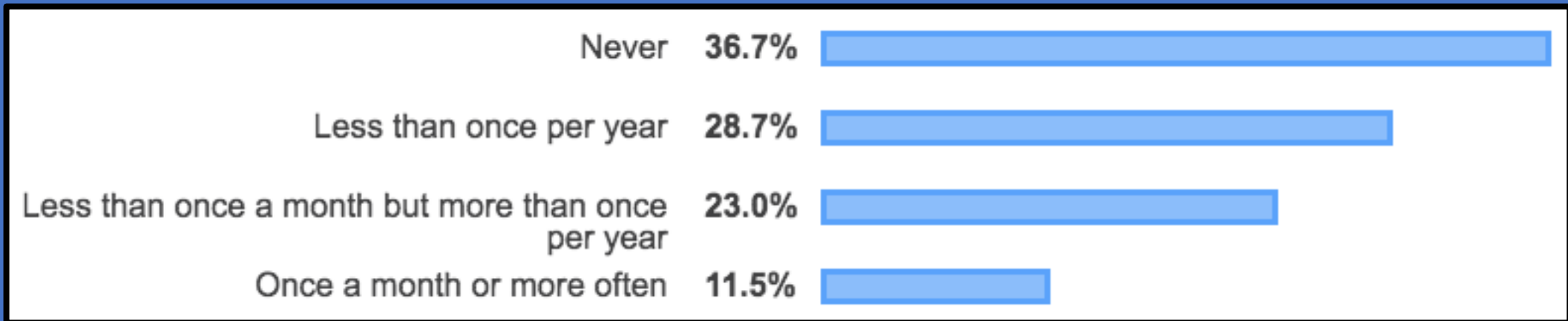
- Do developers review code as part of their work?
 - ✓ Having other people look at your code in order to improve each other's work.



Source: <https://insights.stackoverflow.com/survey/2019>

Open-source projects

- About 65% of professional developers on Stack Overflow contribute to open-source projects once a year or more.



Source: <https://insights.stackoverflow.com/survey/2019>

How much can I make?

Lecture: How much can I make?

Module: The web development career

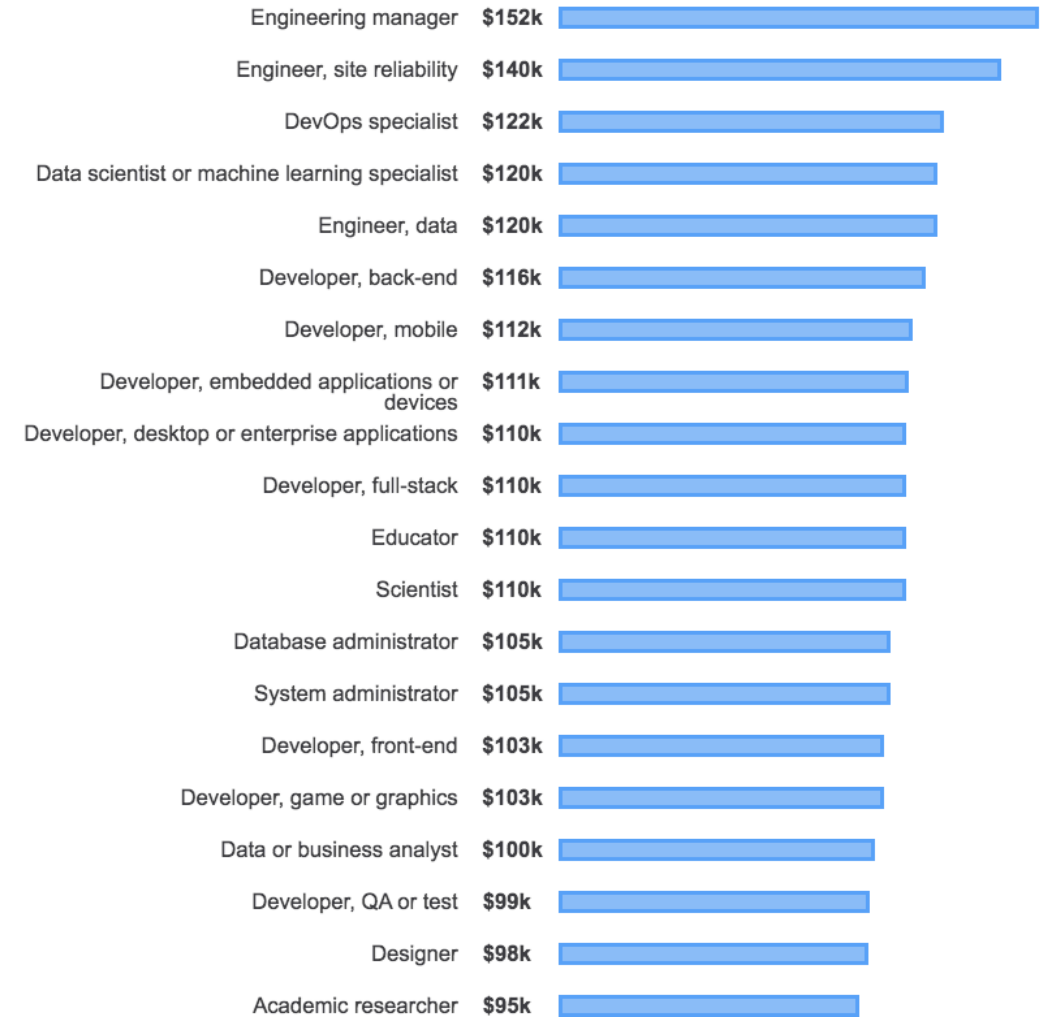
How much can I make?

- Much of your **earning potential** will **vary** depending on **many factors**:
 - ✓ The **size** of the **company**.
 - ✓ The **popularity** of **languages** or **frameworks** that you know.
 - ✓ The **years** of **experience**.
 - ✓ ...
 - ✓ ...



Salary by developer type (US)

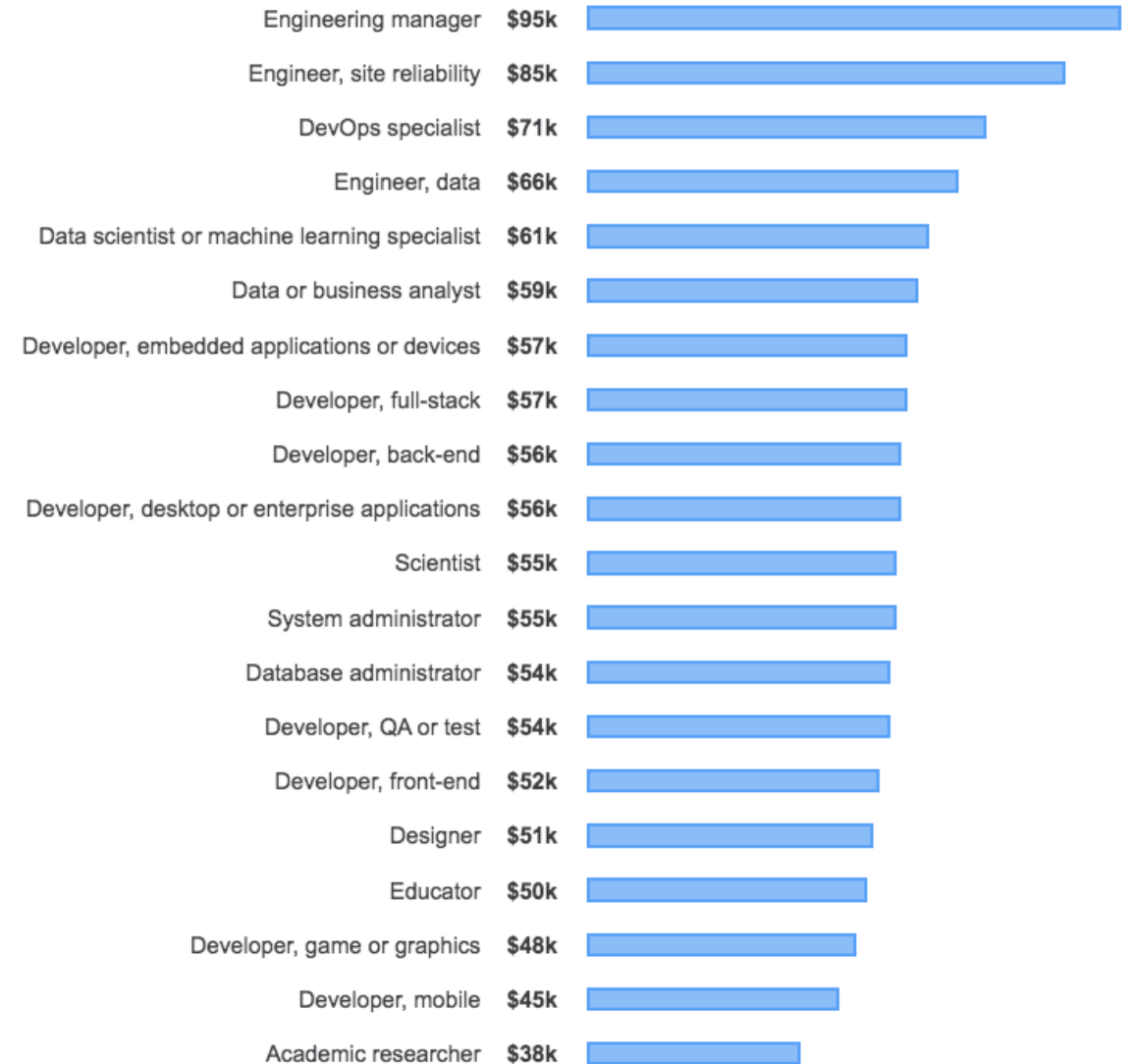
- Engineering managers, SREs, DevOps specialists, and data scientists/engineers command the highest salaries.



Source: <https://insights.stackoverflow.com/survey/2019>

Salary by developer type (worldwide)

- Engineering managers, SREs, DevOps specialists, and data scientists/engineers command the highest salaries.

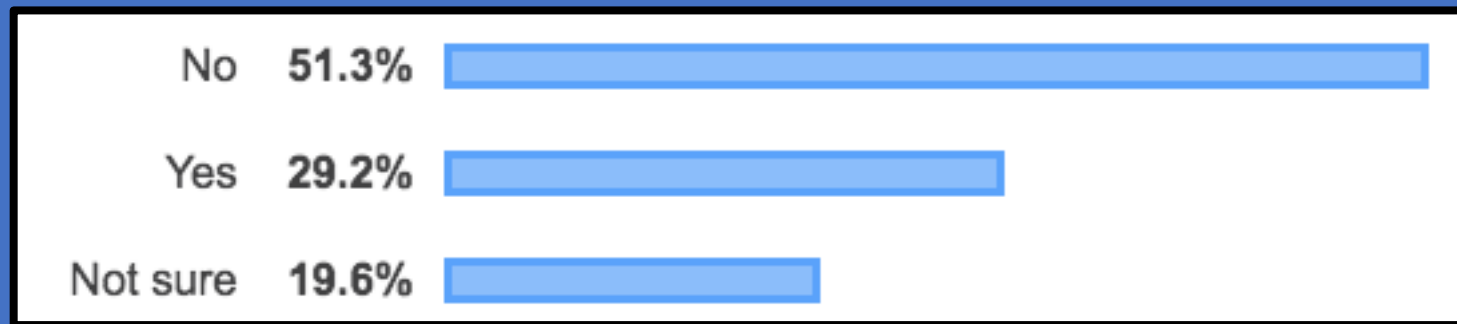


Source: <https://insights.stackoverflow.com/survey/2019>

Back-end development is often considered more challenging when compared to front-end development, and thus reflected in salary.

Do developers need to become managers to make more money?

- Half of respondents don't think they necessarily need to move to people management to continue to grow their salary.



Source: <https://insights.stackoverflow.com/survey/2019>

Education

Education types

Lecture: Education types

Module: Education

Education types

- Education can make a big difference in a web developer career.
- There's four types of education most people take to become web developers:
 1. Instructor-led training
 2. Self-study
 3. Working on your own projects
 4. Real job

1 Instructor led training

Instructor-led training

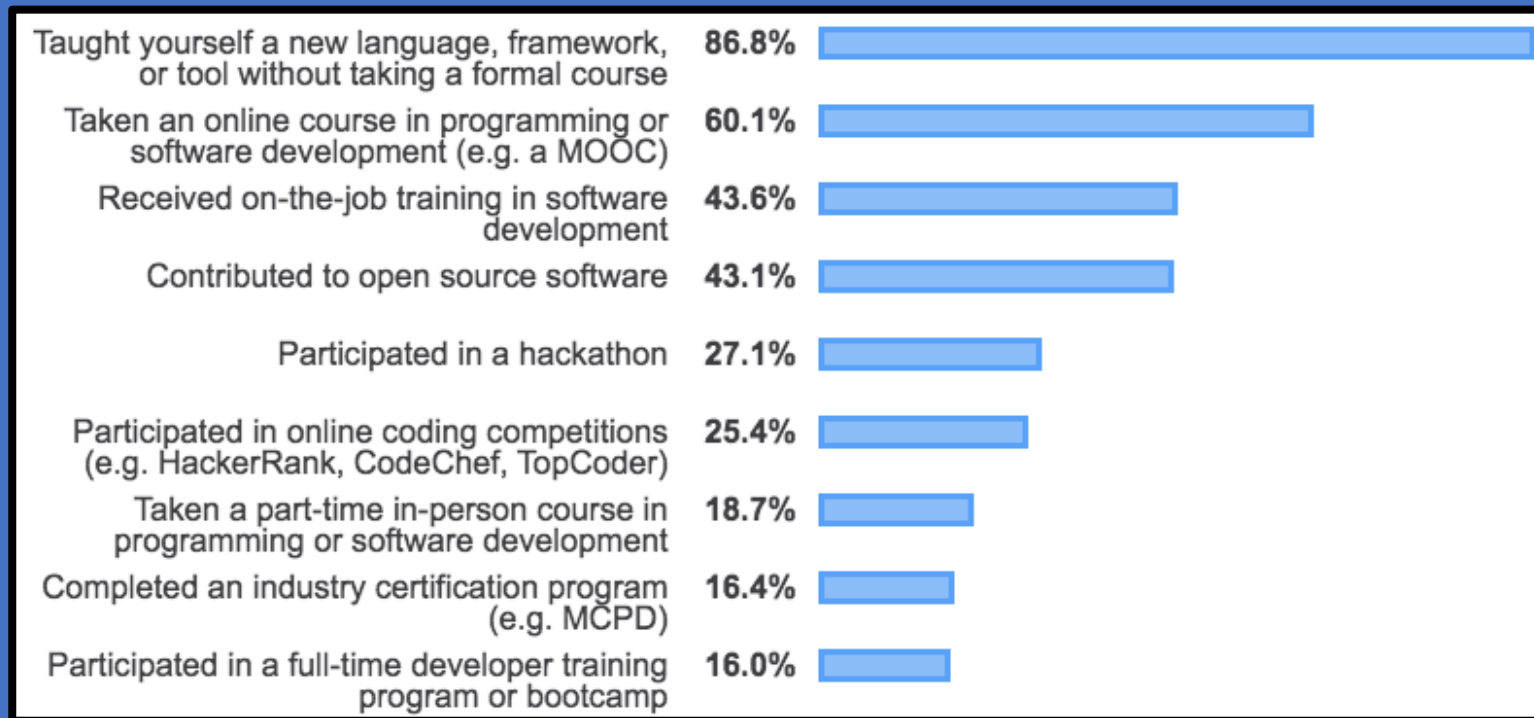
- **Enroll in a in-person course to learn from an instructor.**
 - ✓ The instructor follows a curriculum with exercises, lesson plans, lecture slides, and other tools.
- This type of learning typically happens at a school, university, bootcamp, or in a workshop or conference.
- Works great for someone with less experience who wants to be able to ask questions and work with others.

2 Self-study

**No instructor-led training is going to teach
you everything you need to be successful.**

Lifelong learners

- Developers get involved in **continuous learning**.



Source: <https://insights.stackoverflow.com/survey/2019>

Your **education starts** when you decide to **become a web developer**, but it **never really ends**.

Self-study

- The software industry changes so rapidly that it's not as important to learn specific skills, but rather learn the basics and then adjust quickly to new technologies.
- Some of the keys to becoming a great developer are learning to read online documentation, examining other developers' code, and exploring new ideas that can make you more efficient.

3 Work on projects

No amount of instructor led training, reading books or watching videos online will make you into a great programmer. You'll have to build real projects, including your own projects.

Learning happens as a result of making mistakes with real things. Mistakes make you a stronger developer by improving your ability to find, test, and anticipate errors.

4 Real job

Real job

- **Some learning will only happen** when you're on a **real job**.
- **Working** in a **corporate** or even a **start-up environment** is vastly **different** from the **classroom** or **personal projects**.
- It's **important** to **learn how to work** with **others** in **cross-functional teams** that are **made up of more than just developers**.

Instructor led training: coding bootcamps

Lecture: Instructor led training: coding bootcamps

Module: Education

Bootcamps

- **Bootcamps** can be a **fast way** to **get** the **skills** you need to get into the **developer field** quickly.
 - ✓ They are **designed** to **teach** the **practical skills** necessary to **get** a **job** in the **industry** in a **condensed timeframe**.
- **Bootcamps** are usually **full-time, in-person, immersive experience**.

2021 Best Coding Bootcamps

52 Top Schools: Rankings, Reviews, and Courses

Not Sure Where to Apply?

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By: **The SwitchUp Team**

Last Updated: September 21, 2020



Despite recent economic downturns, hiring managers at major tech companies and startups alike are still looking for highly skilled coding professionals. A [whopping 84%](#) of them agree that bootcamp grads are likely to be top performers and are just as prepared to succeed as their college-degree holding counterparts—if not more so. Clearly, intensive coding bootcamps that equip participants with in-demand skills are well worth the investment of time, energy, and financial resources.

SwitchUp has been tracking coding bootcamps since 2014, and we've become a trusted source for helping people identify the best choices in coding bootcamps. Only those bootcamps whose coding programs have been thoroughly reviewed by at least 20 students and earned a minimum 4 out of 5 stars have made it onto our latest ranked list, below. Click [here](#) to see our full methodology or read on for more information about [coding education and careers](#).

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★★★★★ 5.0/5 (based on 54 reviews of Coding Programs)

Locations: Barcelona, Madrid**Courses:** Web Development Bootcamp with Javascript - Barcelona, Web Development Bootcamp with Javascript - Madrid

<https://www.switchup.org/rankings/best-coding-bootcamps>

Some campuses are now running hybrid classes: learn to code from home or on campus. [Learn more >](#)



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9 weeks

Web Development Part-Time

24 weeks

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Data Science Full-Time

9 weeks

Data Science Part-Time

24 weeks

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Bootcamps

- Many bootcamps run 40 hours or more per week.
- This hard work often pays off, since they teach skills not often taught in universities, like pair-programming, test-driven development, continuous integration, and cutting-edge tools.

Disadvantages of Bootcamps

- Many bootcamps have pretty low acceptance rates compared to traditional schools.
 - ✓ They pick and choose the students with the highest chance of success.
 - ✓ Not everyone is prepared for 10 to 12 weeks of intensive training, especially if you have a current job and have no other means of support.

Disadvantages of Bootcamps

- Too much **practical knowledge without an understanding of why you should do something**, can also be a **disadvantage**.
 - ✓ **Not everyone knows exactly what they want to do when they first go to college.**
 - ✓ **Only about 50% of college graduates have jobs that relate to their degree.**
 - ✓ **Over 30% of college educated workers never find jobs related to their major.**
- **Expensive tuition**
 - ✓ **Some of them costs more than 10,000 US\$ for 3 weeks training !!!**

Bootcamps for college graduates

- **Majority of bootcamp attendees are college graduates.**
 - ✓ Hence, **most students** are already **prepared** for the type of **intensive training** required in a **bootcamp**.
- **Bootcamps can be great if you're sure that being a web developer is what you want to do.**
 - ✓ **Committing** to an **immersive program**, and the **financial challenges**, are not for everyone.

Instructor led training: traditional education

Lecture: Instructor led training: Traditional education

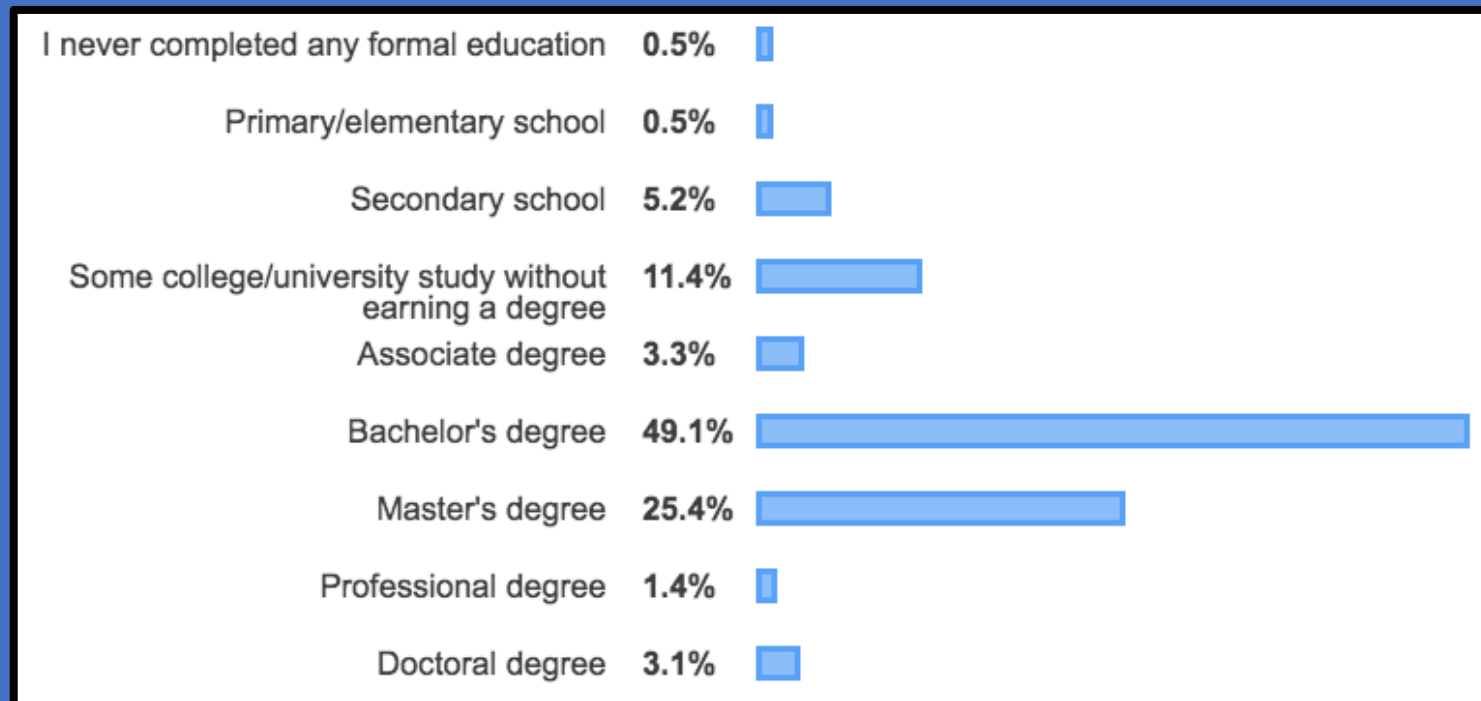
Module: Education

Undergraduate education

- **Higher education** is an **important** part of **many** careers, but **not everyone thrives** in a **standard educational environment**.
- Do you **really need** to **get a college degree** to **get a developer job**?
 - ✓ People like **Steve Jobs** and **Mark Zuckerberg** **never finished** their **college education**.
- A **better question** is **what value** does having a **college degree** **bring** to my **ability** to **get hired**?

Undergraduate education worldwide

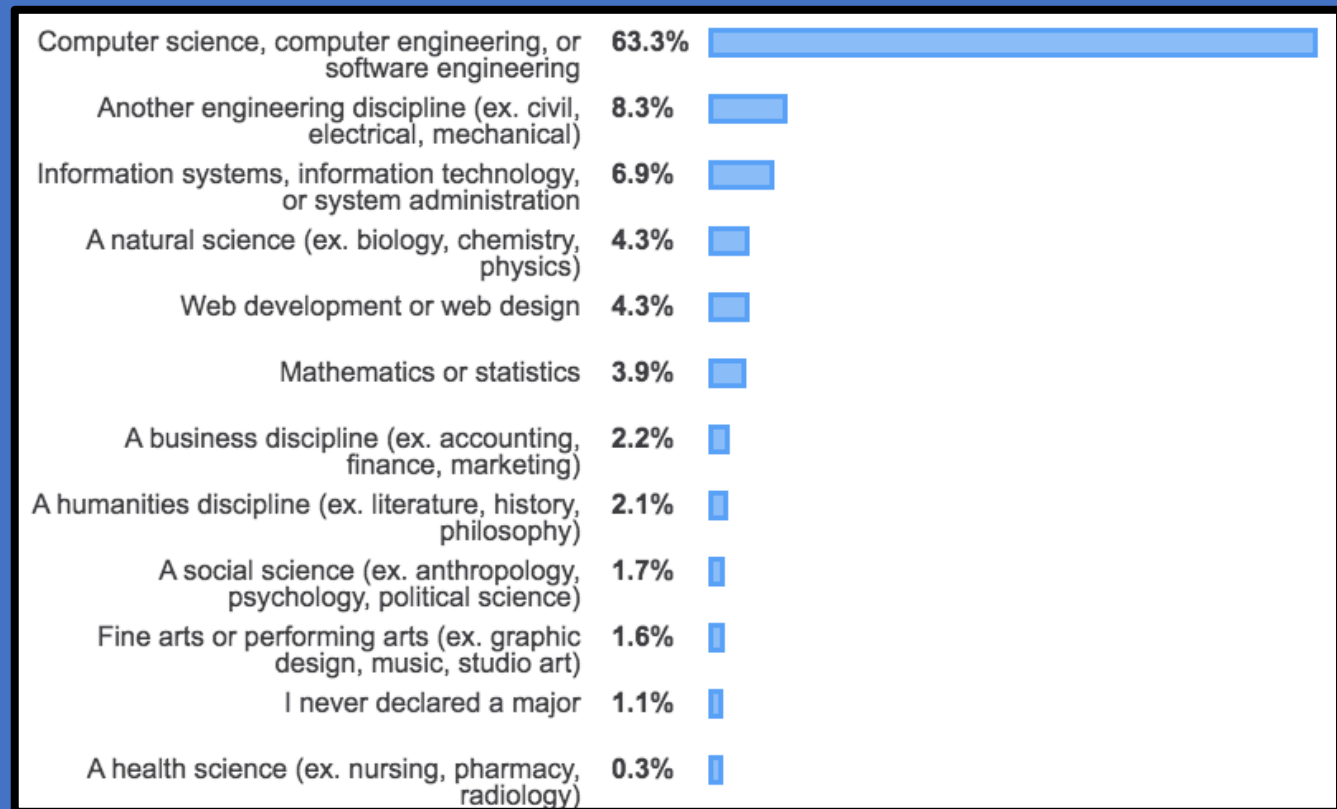
- About **three-fourths** of professional developer respondents have the equivalent of a **bachelor's degree or higher**.



Source: <https://insights.stackoverflow.com/survey/2019>

Undergraduate education worldwide

- Over 60% of the professional developers who studied at the university level, said they majored in computer science, computer engineering, or software engineering.



Source: <https://insights.stackoverflow.com/survey/2019>

The cost of a college education in US

- The **average cost of attending a private university** is nearly **\$50,000 annually**.
 - ✓ **Three times the price since 1990.**
 - ✓ **Tuition** has climbed at a **faster rate than inflation**, and **median income levels**.
- However, having a **college education** does **make a big difference**.
 - ✓ **College graduates** earn about **\$1 million more** over their **lifetime** than **people without a degree**.

College degree

- **Some companies require a college degree in order to be hired.**
 - ✓ About 65% of all jobs require a post-secondary education.
- **However, in recent years, some important companies stopped requiring a college degree:**
 - ✓ Apple
 - ✓ Google
 - ✓ IBM
 - ✓ ...

Benefits of a college education

- College students learn to solve problems and explore research into higher level concepts.
- General education gives students a firm and wide foundation.
 - ✓ Math and physics courses can be very useful to computer science
 - ✓ Students can also be exposed to business classes that will teach them how to write proposals and documents for decision makers.

When comparing a traditional college degree to a self-study or bootcamp, a college degree has advantages in terms of earning potential, learning methodologies, and increasing your ability to get hired.

Self-study: online learning

Lecture: Self-study: online learning

Module: Education

Online education

- If you're a **self starter**, your **education** can happen **completely online**.
- **Online learning** has several **advantages**
 - ✓ You can **learn where** and **when** you like.
 - ✓ It's **low cost**, and you have **access** to an **enormous amount** of **content**.

Learning from Google and Youtube

- **Google and YouTube are great ways to find quick answers.**
 - ✓ but they can be **full of good and bad** as well as **old** and sometimes **distracting answers**.



Google Search

I'm Feeling Lucky

Google offered in: [Português \(Brasil\)](#)

<https://google.com>



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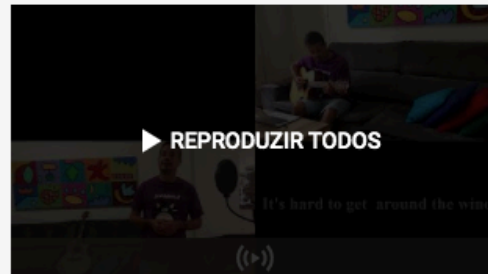
Biblioteca

Tudo Mixes Música Guitarras elétricas violão acústico Indie rock Os Beatles New Wave Red Hot Chili Peppers Blues-rock Álbums >



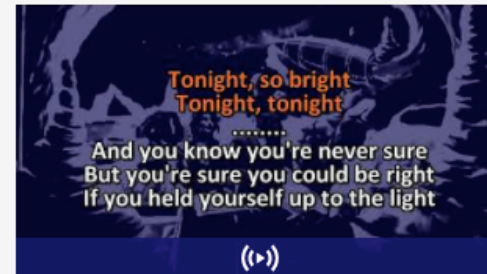
Pearl Jam / Yellow Ledbetter / acoustic cover

Justin Comolli
3,5 mil visualizações • há 2 meses



Meu mix

Uma playlist personalizada para você



Mix de Smashing Pumpkins - Tonight, Tonight - Best Version - Karaoke...

The Smashing Pumpkins, Red Hot Chili Peppers, R.E.M.e outros



Indie / Bedroom / Pop / Surf Rock - 24/7 Radio - Nice Guys...

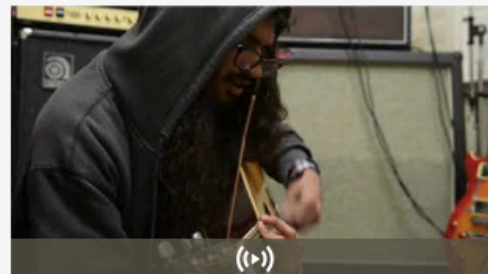
Nice Guys
371 assistindo

AO VIVO AGORA



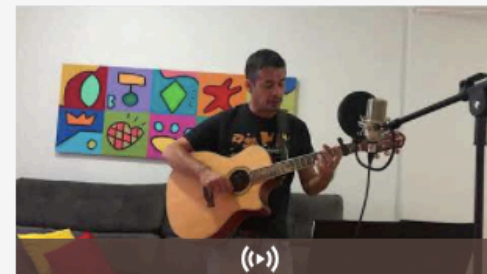
"Essa menina toca melhor do que eu". Convite para worksho...

Kiko Loureiro ✓
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Mix de my bloody valentine - "when you sleep" (acoustic cover)

Elliott Smith, Slowdive, My Bloody Valentinee outros



Mix de Pink moon - Nick Drake acoustic cover

Nick Drake, Pink Floyd, John Lennone outros



Mix de Purple Rain (Prince)- Acoustic Cover (+Tabs & Tutorial)

Prince, Michael Jackson, Pink Floyde outros

<https://youtube.com>



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monthly visitors to Stack

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Times a developer got help

179% ROI

from companies using Stack Overflow for Teams

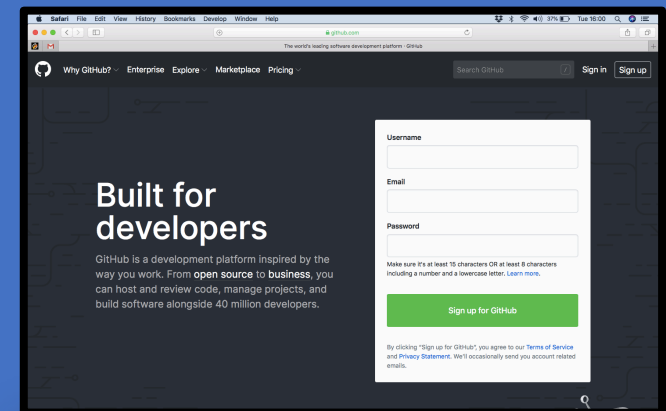
5,000+

Stack Overflow for Teams instances active every day

<https://stackoverflow.com>

Learning by forking a real project

- Sometimes there is **no substitute** for **working** on a **real project**.
- **Full code** for many major **projects** is **freely available** at sites like **GitHub**.
 - ✓ This is called **forking** and **allows** you to **create** your **own version** of an **existing project**.
- You can **learn a lot** just from **reading** how **someone else solved** the **problem** in a **real project**.





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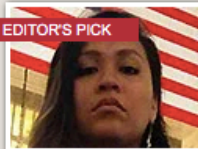
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Learning path for a web developer

Lecture: Learning path for a web developer

Module: Education

HTML is the language that front-end, back-end and full-stack developers must to know.

Typical learning path for front-end developers

1. HTML
2. CSS
3. JavaScript
4. Content Management System - CMS (e.g. WordPress)
5. Front-end libraries and frameworks:
 - ✓ jQuery
 - ✓ Bootstrap
 - ✓ Angular
 - ✓ React.js
 - ✓ Vue.js
 - ✓ ...



Back-end architecture concepts and patterns are technology agnostic, but a learning path for back-end developers is usually programming language dependent.

Python

1. HTML
2. Python
3. Flask
4. Django
5. WordPress
6. ...

JavaScript

1. HTML
2. JavaScript
3. Node.JS
4. WordPress
5. ...

PHP

1. HTML
2. PHP
3. Laravel
4. WordPress
5. ...

Typical learning paths for back-end developers

Java

1. HTML
2. Java SE
3. Jakarta EE
4. Spring
5. Spring Boot
6. WordPress
7. ...

Ruby

1. HTML
2. Ruby
3. Ruby on Rails
4. WordPress
5. ...

C#

1. HTML
2. .NET,
3. ASP.NET
4. WordPress
5.

Learning path for a back-end developer

- **New back-end developers need to study at least:**
 - ✓ One structured database language.
 - ✓ One NoSQL language and
 - ✓ One language for merging data with templates.

Learning path for a back-end developer

- **WordPress** is so **popular** that it powers about **35%** of the **web**.
 - ✓ Hence you should **at least know** how to **set it up**.
- **Entire businesses** are **built** around the **WordPress stack**.

Typical learning path for a full-stack developer:

Front-end learning path + Back-end learning path



**Web development is a career
path without end!**

Conclusion

References

Lecture: References
Module: Conclusion



Developer Survey Results 2019



Overview

Key Results

Developer Profile

Technology

Work

Community

Methodology

Back to top



Overview

This year, nearly 90,000 developers told us how they learn and level up, which tools they're using, and what they want.

Stack Overflow's annual Developer Survey is the largest and most comprehensive survey of people who code around the world. Each year, we field a survey covering everything from developers' favorite technologies to their job preferences. This year marks the ninth year we've published our annual Developer Survey results, and nearly 90,000 developers took the 20-minute survey earlier this year.

Despite our survey's broad reach and capacity for informing valuable conclusions, we acknowledge that

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<https://insights.stackoverflow.com/survey/2019>

Questions?